

Calculus makes motion understandable

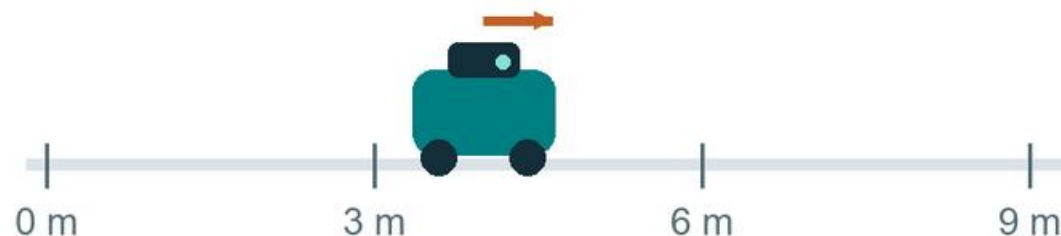
A visual introduction for incoming master's students

How fast?
How much?
What happens next?

45 minutes · intuition first

Graphs, short examples, and robot motion

A robot moves. Calculus explains how.



$t = 2.0 \text{ s}$

$x = 4.0 \text{ m}$

$v = 4.0 \text{ m/s}$

Model: $x(t) = t^2$ • positive direction $\rightarrow$

Today: interpret change, accumulate it, and follow it through a robot.

Two questions unlock a changing world

Calculus connects a changing rate with an accumulated amount.

DIFFERENTIATION

**How fast is it
changing now?**

Tank level → filling rate

Temperature → heating rate

Position → velocity

INTEGRATION

**How much has
accumulated?**

Flow rate → water added

Power → energy used

Velocity → displacement

A derivative gives a local rate. An integral adds contributions over an interval.

Where you will use it in autonomous systems

It helps you model, predict, and improve behaviour.

MOTION & CONTROL

Move smoothly

Position → velocity → acceleration

Plan turns, braking, and actuator demands.

STATE ESTIMATION

Track motion

Combine rates over time.

Use wheel and inertial data to estimate movement.

OPTIMISATION

Reduce error

Find how a cost changes.

Tune a trajectory, a calibration, or a learned model.

The benefit: understand what an algorithm computes, and recognise when its model fails.

From areas and tangents to a language of motion

Many people contributed; Newton and Leibniz brought key ideas together.

ANCIENT FOUNDATIONS

Archimedes

Approximated curved areas and volumes with increasingly fine pieces.

1660S–1670S

Newton & Leibniz

Developed calculus independently, building on earlier ideas.

LATER FOUNDATIONS

Limits & rigour

Later mathematicians clarified why the methods work.

Today: models and computation.

The original questions are still ours: how does something change, and how much does it add up?

A function is a relationship between quantities

Read $x(t)$ as “position x at time t ”.

Input $\rightarrow$ rule $\rightarrow$ output

t : time, in seconds

$x(t)$: position, in metres

$v(t)$: velocity, in metres/second

A graph shows how the output changes as the input changes.

A robot moves. Calculus explains how.



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Model: $x(t) = t^2$ • positive direction $\rightarrow$

For the motion examples, use one straight line and a fixed positive direction.

Velocity is the rate of change of position

Start with a finite interval, then ask what happens near one instant.

OVER AN INTERVAL

Average velocity

$$\Delta x / \Delta t$$

Example: 6 m in 3 s $\rightarrow$ 2 m/s

Δ means “change in”.

AT ONE INSTANT

Instantaneous velocity

$$v(t) = dx/dt$$

Use smaller and smaller time intervals around that instant.

Geometrically: the tangent slope.

Differentiate position with respect to time to obtain velocity.

Zoom in: an average becomes an instant

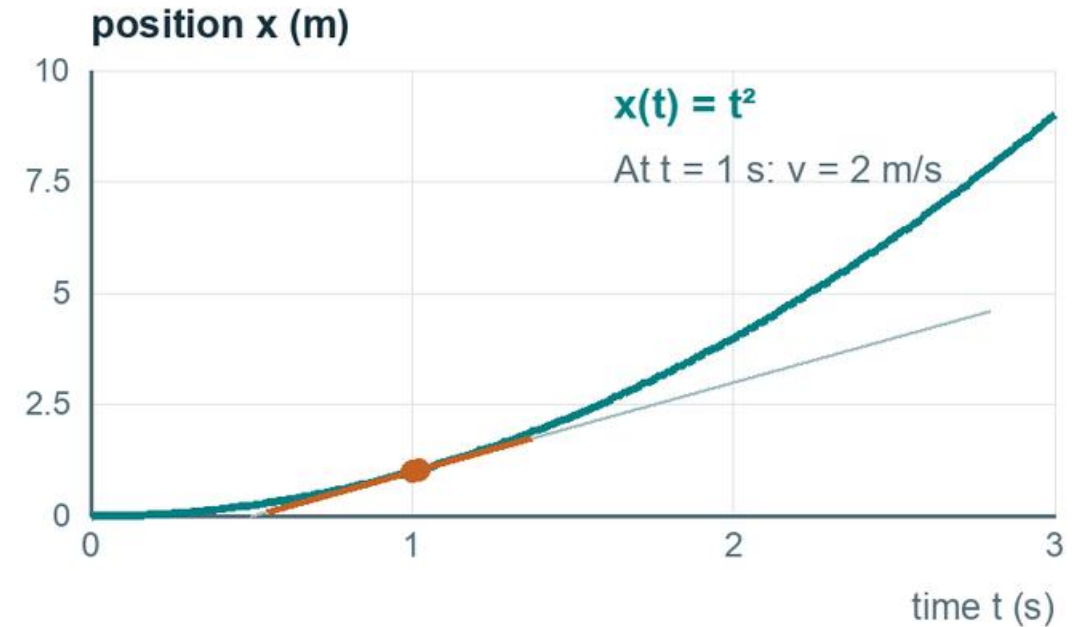
The second point approaches the first; the secant approaches the tangent.

For $x(t) = t^2$ at $t = 1$ s

Average slope = $2 + \Delta t$
As $\Delta t \rightarrow 0$, the slope $\rightarrow 2$

$$v(t) = \lim_{\Delta t \rightarrow 0} \frac{x(t + \Delta t) - x(t)}{\Delta t}$$

$\Delta t = 0.02$ s average $v = 2.02$ m/s



ANIMATED DIAGRAM

At $t = 1$ s, the robot's instantaneous velocity is 2 m/s.

One motion, three useful descriptions

A deliberately simple model: $x(t) = (1 \text{ m/s}^2)t^2$.

POSITION

$$x(t) = t^2$$

Where is it?

At $t = 2 \text{ s}$:
 $x = 4 \text{ m}$

VELOCITY

$$v(t) = 2t$$

How fast does position
change?

At $t = 2 \text{ s}$:
 $v = 4 \text{ m/s}$

ACCELERATION

$$a(t) = 2$$

How fast does velocity
change?

At $t = 2 \text{ s}$:
 $a = 2 \text{ m/s}^2$

$x \rightarrow \text{differentiate} \rightarrow v \rightarrow \text{differentiate} \rightarrow a$ | Units change at every step.

Read the motion without calculating

On this position–time graph, what is the sign of velocity at A, B, and C?

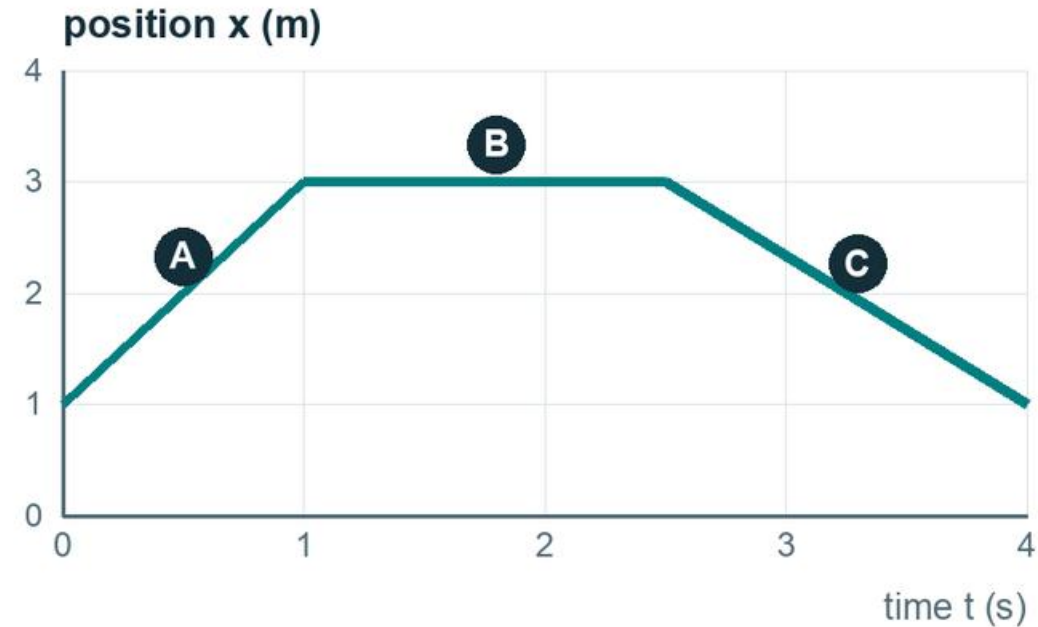
A: positive, zero, or negative?

B: positive, zero, or negative?

C: positive, zero, or negative?

Bonus: does a high position automatically mean a high velocity?

Same position. Three different rates of change.



Look at the slope of the graph, not its height.

A few rules handle many simple models

Here u is a scalar variable, c is constant, and angles are in radians.

Constant

$$d(c)/du = 0$$

A fixed quantity has no change.

Power

$$d(u^n)/du = n u^{n-1}$$

For the polynomial examples here: $u^2 \rightarrow 2u$.

Scale & sum

$$(c f + g)' = c f' + g'$$

Differentiate each term, then combine.

Sine & cosine

$$(\sin u)' = \cos u; (\cos u)' = -\sin u$$

Useful when a robot rotates.

Example: $d(3u^2 + 2u + 5)/du = 6u + 2$. The chain rule comes later.

Add tiny pieces to recover a total

Over a short interval: displacement $\approx$ velocity $\times$ elapsed time.

One strip: $v(t) \Delta t$

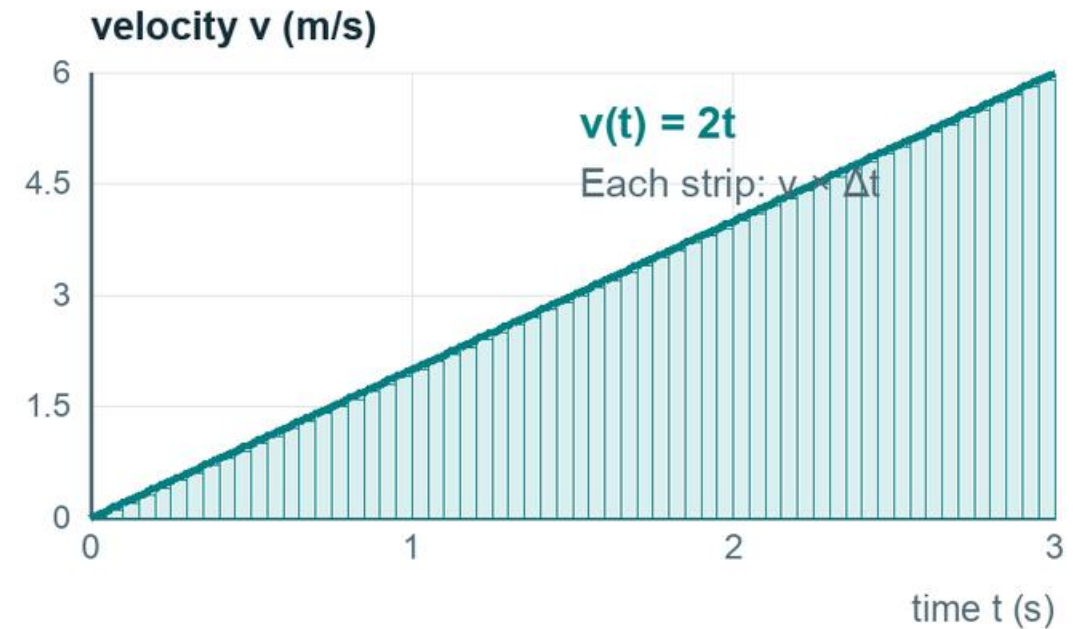
Many strips: $\sum v[i] \Delta t$

An integral is the limiting sum as the time slices become arbitrarily fine.

$$\Delta x = \int v(t) dt$$

over the time interval

60 slices total ≈ 8.85 m exact = 9 m



ANIMATED DIAGRAM

The units explain the answer: $(\text{m/s}) \times \text{s} = \text{m}$.

How far does the robot move in three seconds?

Use the same motion: $v(t) = 2t$, with t in seconds and v in m/s.

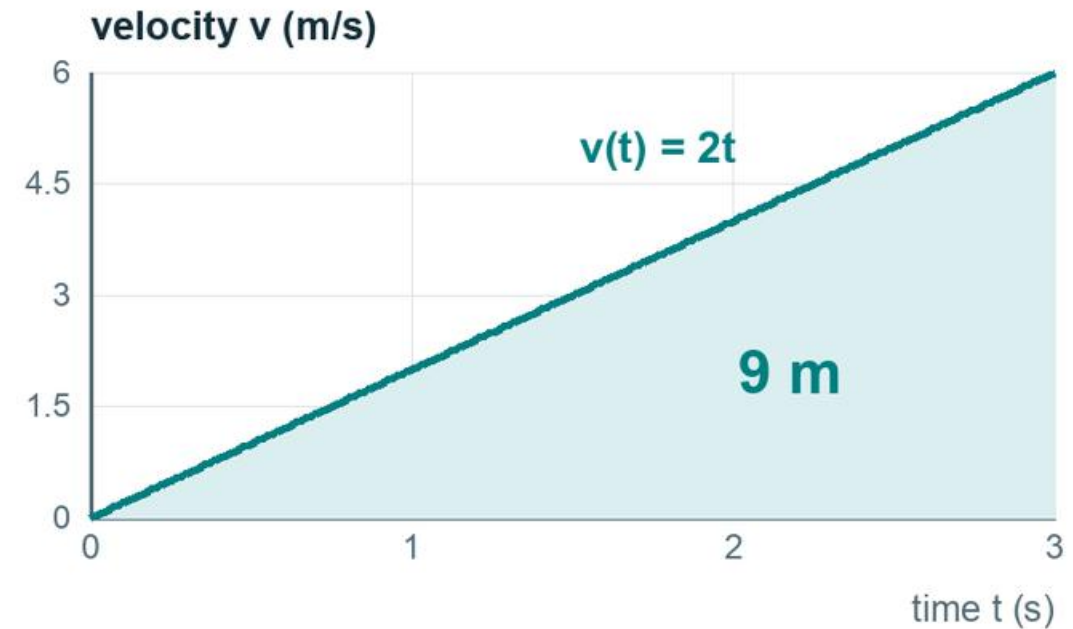
From $t = 0$ s to $t = 3$ s

Triangle area

$$\frac{1}{2} \times 3 \text{ s} \times 6 \text{ m/s} = 9 \text{ m}$$

$$\int_0^3 2t \, dt = [t^2]_0^3 \\ = 9 - 0 = 9 \text{ m}$$

Exact accumulation from 0 s to 3 s



If $x(0) = 0$ m, then $x(3) = 9$ m. If $x(0) = 5$ m, then $x(3) = 14$ m.

Displacement and distance can disagree

A robot travels at $+2$ m/s for 2 s, then at -1 m/s for 1 s.

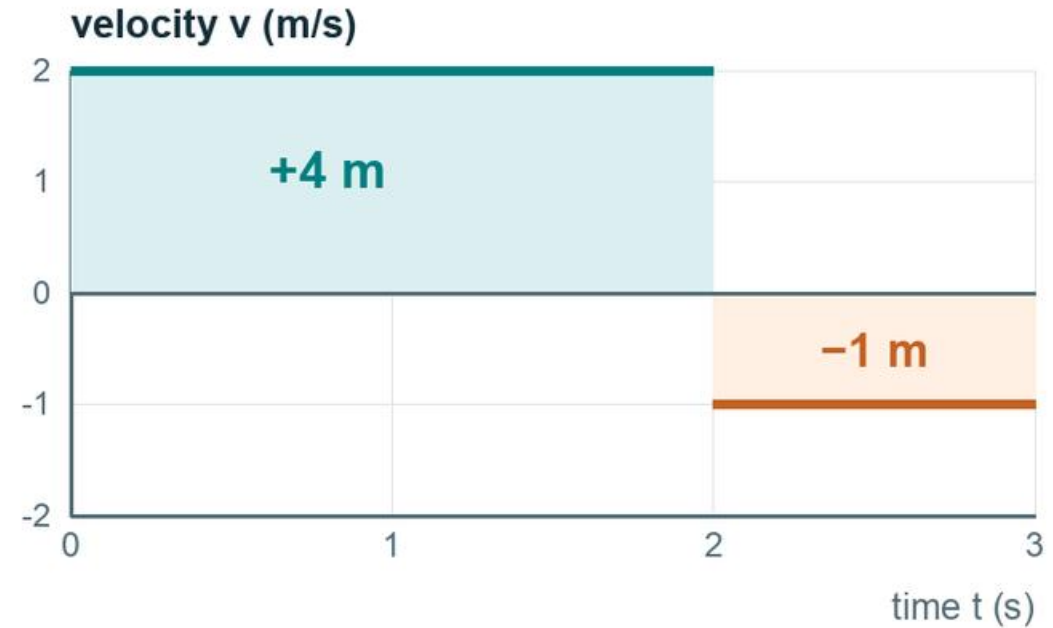
What is the final change in position?

How far did it travel in total?

Signed velocity $\rightarrow$ displacement

Speed $|v| \rightarrow$ distance travelled

Forward, then backward



Areas below zero subtract from displacement; they still add to distance travelled.

Differentiation and integration are connected

The fundamental theorem of calculus makes the connection precise.

FROM A QUANTITY TO ITS RATE

Differentiate

$$x(t) \rightarrow dx/dt = v(t)$$

The initial position disappears when you differentiate.

FROM A RATE TO A QUANTITY

Integrate + initial value

$$x(t) = x(0) + \int_0^t v(\tau) d\tau$$

Add the starting position.

An indefinite integral gives a family: $\int 2t dt = t^2 + C$. The initial value fixes C .

A computer works with small time steps

Calculus supplies the model; numerical methods approximate it.

ESTIMATE A DERIVATIVE

Difference / time step

$$v[k] \approx (x[k] - x[k-1]) / \Delta t$$

Tiny measurement errors can become large rate errors.

APPROXIMATE AN INTEGRAL

Sum rate $\times$ time step

$$x[k+1] \approx x[k] + v[k] \Delta t$$

A constant velocity bias creates a position error that grows with time.

Practical habits: check units, sample times, sensor noise, and initial conditions.

What if the unknown is the whole motion?

A differential equation relates a quantity to its derivatives.

$$m \, dv/dt = F - b \, v$$

m : mass

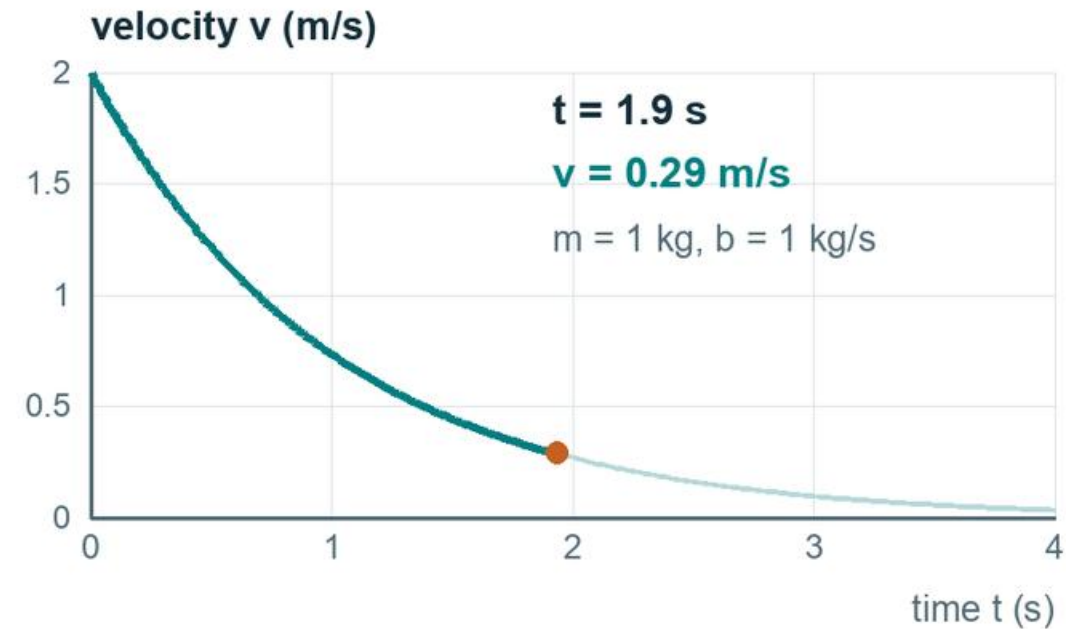
F : driving force

bv : a simple drag force

The rule gives acceleration.

With an initial velocity, it predicts how velocity evolves.

Drag slows the robot continuously



ANIMATED DIAGRAM

A model of change + an initial condition → a prediction over time.

Same law, different starting conditions

Without driving force: $dv/dt = -k v$, where $k = b/m > 0$.

One possible solution form

$$v(t) = v(0) e^{-kt}$$

$v(0)$ chooses the initial velocity.
 $1/k$ sets a characteristic time scale.

What you should recognise

The equation: the rule of change

The initial condition: the start

The solution: the predicted motion

For $k = 1 \text{ s}^{-1}$, after one second the velocity is about 37% of its initial value.

A change often passes through another quantity

Time changes a joint angle. The joint angle changes the tip's position.

INPUT

Time t

Seconds pass.

The joint rotates.

INTERMEDIATE

Angle $\theta(t)$

The angle depends on time.

How fast does it rotate?

OUTPUT

Position $x(\theta)$

The tip position depends on the angle.

How sensitive is x to θ ?

$$dx/dt = (dx/d\theta) \times (d\theta/dt) \mid \text{sensitivity to angle} \times \text{angular velocity}$$

Differentiate the outside, then the inside

For a dimensionless scalar input u , let $y = (3u + 1)^2$.

$$u \rightarrow z = 3u + 1 \rightarrow y = z^2$$

Outside: $dy/dz = 2z$

Inside: $dz/du = 3$

Multiply: $dy/du = 2z \times 3$

$$dy/du = 6(3u + 1)$$

At $u = 1$: the local rate is 24.

A small Δu gives $\Delta y \approx 24 \Delta u$.

For $y = f(g(u))$: $dy/du = f'(g(u)) g'(u)$. Remember the inner derivative.

Joint speed becomes tip velocity

One rigid link of length L ; θ is measured from the positive horizontal axis.

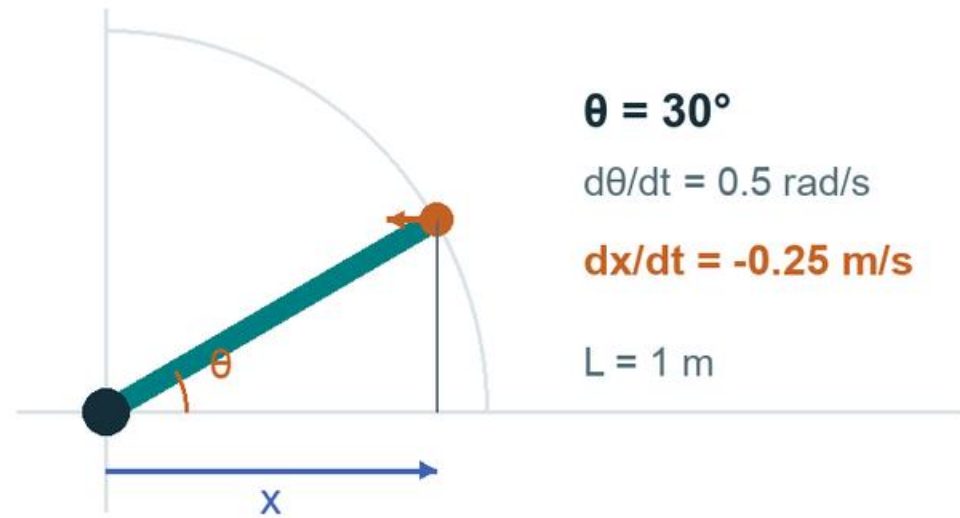
$$x(t) = L \cos(\theta(t))$$

$$dx/dt = -L \sin(\theta) \times d\theta/dt$$

$$L = 1 \text{ m}, \theta = 30^\circ = \pi/6 \text{ rad}$$
$$d\theta/dt = 0.5 \text{ rad/s}$$

$$dx/dt = -0.25 \text{ m/s}$$

Constant joint speed $\neq$ constant horizontal speed



The arrow shows horizontal velocity only.

ANIMATED DIAGRAM

The negative sign means the tip moves left in x . Its full velocity also has a vertical component.

The same idea scales to many dependencies

A short look ahead: you do not need the matrix details today.

ROBOT WITH MANY JOINTS

Joint rates $\rightarrow$ tip velocity

Tip position depends on several angles.

Add each joint's contribution.

The Jacobian collects these sensitivities.

LEARNING & OPTIMISATION

Parameter $\rightarrow$ prediction $\rightarrow$ loss

A model's error depends on its output, which depends on its parameters.

Repeated chain rules carry sensitivity through the computation.

The chain rule connects local effects across a complete system.

Which rate is missing?

A robot joint is at $\theta = 60^\circ$ and rotates at $d\theta/dt = 0.2$ rad/s. Use $L = 1$ m.

$$x = L \cos \theta$$

$$dx/dt = -L \sin \theta \times \underline{\hspace{2cm}}$$

1. Fill in the missing factor.

2. Should dx/dt be positive or negative?

3. Estimate dx/dt .

Use $\sin 60^\circ \approx 0.866$.

Explain the sign in words before using a calculator.

Turn a robot question into a calculus question

Choose the quantity you need, then follow the units and dependencies.

“How fast is the robot moving?”

Differentiate position

Velocity estimate

“Where will it be after this interval?”

Integrate velocity + initial position

Predicted position

“How will it react to force?”

Use a differential equation

Simulated motion

“How do joint commands move the tip?”

Apply the chain rule

Tip velocity

“How can I reduce a model’s error?”

Differentiate the objective

Local update direction

Start every model with a sketch, a coordinate convention, units, and assumptions.

TAKE THESE FOUR IDEAS WITH YOU

Calculus is a toolkit for understanding change

You now have the vocabulary for motion, estimation, control, and learning.

- | | | |
|----|------------------------------|--|
| 01 | Derivative | How fast does it change here? |
| 02 | Integral | How much accumulates over an interval? |
| 03 | Differential equation | What rule governs the change? |
| 04 | Chain rule | How does change pass through dependencies? |

When you see a moving robot, ask: what changes, at what rate, and what does it depend on?